TURNING UP THE HEAT: MIN-p SAMPLING FOR CREATIVE AND COHERENT LLM OUTPUTS

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ABSTRACT

Large Language Models (LLMs) generate text by sampling the next token from a probability distribution over the vocabulary at each decoding step. Popular sampling methods like top-p (nucleus sampling) often struggle to balance quality and diversity, especially at higher temperatures which lead to incoherent or repetitive outputs. We propose **min-p sampling**, a dynamic truncation method that adjusts the sampling threshold based on the model's confidence by using the top token's probability as a scaling factor. Our experiments on benchmarks including GPQA, GSM8K, and AlpacaEval Creative Writing show that min-p sampling improves both the quality and diversity of generated text across different model families (Mistral and Llama 3) and model sizes (1B to 123B parameters), especially at higher temperatures. Human evaluations show a preference for min-p, in both text quality and creativity. Min-p sampling has been adopted by popular open-source LLM frameworks, including Hugging Face Transformers, VLLM, and many others, highlighting its considerable impact on improving text generation quality.

1 INTRODUCTION

A central challenge for Large Language Models (LLMs) is the trade-off between creativity and coherence during text generation. Popular sampling strategies like top-p sampling (Holtzman et al., 2020) and temperature scaling (Ackley et al., 1985) are widely adopted, but increasing temperature to enhance diversity often reduces coherence, while more conservative sampling limits creativity.

We address the *quality-diversity tradeoff* with **min-p sampling**, a dynamic truncation method that adjusts the sampling threshold based on model confidence using the top token's probability as a scaling factor. This approach includes diverse options when the model is uncertain and keeps only high-confidence tokens when confident, balancing creativity and coherence even at high temperature.

Our experiments on benchmarks including **GPQA** (Rein et al., 2023), **GSM8K** (Cobbe et al., 2021), and **AlpacaEval Creative Writing** (Li et al., 2023b). Our results show that min-p sampling outperforms top-p and other popular decoding methods such as top-k (Fan et al., 2018) and η-sampling (Hewitt et al., 2022a) by maintaining coherence while allowing for increased diversity, particularly at high-temperatures. We also conducted a comprehensive **human evaluation** to compare the quality and diversity of text generated by min-p with that generated by traditional sampling methods. The results indicate a clear preference for min-p, with participants rating min-p outputs as superior in quality and diversity.

The contributions of this paper are as follows:

- We introduce **min-p sampling**, a novel dynamic truncation method that effectively balances creativity and coherence in LLM-generated text, particularly at high temperatures.
- We present comprehensive **experimental results** across benchmarks, model families and sizes, demonstrating that in practice min-p consistently improves quality and diversity of generated text, especially at higher temperatures, compared to sampling methods like top-p.

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- We validate the practical utility of **min-p** through an extensive **human evaluation**, showing instances where human evaluators prefer min-p outputs over those generated by other methods in terms of both quality and diversity.
- We provide practical **empirical guidelines** for hyperparameter selection and usecases.
- The rapid **adoption of min-p** by the open-source LLM community highlights its practical benefits, even for large reasoning models such as Deepseek R1 (DeepSeek-AI et al., 2025).

By introducing min-p and offering empirical guidelines for its use, we enable exploration of high-temperature creative text generation without sacrificing coherence. Our results demonstrate that min-p is a viable and superior alternative to existing sampling methods, both at standard and high-temperature settings, respectively, marking a major contribution to generative language modeling.

2 Related Work

We review current sampling methods and quality-diversity tradeoff, establishing motivation for min-p.

Greedy Decoding and Beam Search. These deterministic decoding strategies select the highest probability token at each step (Freitag & Al-Onaizan, 2017), which often produces repetitive and generic text due to lack of diversity. Beam search also incurs a notable runtime performance penalty.

Stochastic Sampling Methods. These methods inject diversity via randomness in token selection. **Temperature scaling** adjusts distribution sharpness to balance diversity and coherence (Ackley et al., 1985); however, higher temperatures often give incoherent outputs, limiting applicability. **Top-k sampling** selects from the k most probable tokens (Fan et al., 2018), offering a simple way to prevent sampling unlikely tokens, but doesn't dynamically adapt to varying confidence levels across contexts.

Top-p sampling (nucleus sampling) restricts the token pool to those whose cumulative probability exceeds a total threshold p (Holtzman et al., 2020), balancing quality and diversity by preserving the "nucleus" of high-probability tokens. However, at higher temperatures, top-p can still allow low-probability tokens into the sampling pool, leading to incoherent outputs. This specific trade-off between creativity and coherence at high temperatures is a key limitation we address with min-p.

Entropy-Based Sampling. Recent approaches like **(η-sampling)** and **mirostat sampling** dynamically adjust the sampling pool based on token distribution entropy (Hewitt et al., 2022a; Basu et al., 2021). While these show promise, they often require complex parameter tuning and can be computationally expensive. We detail our experimental challenges running η sampling in Appendix A.3.

Dynamic Threshold Methods. Recent work has explored adaptive threshold mechanisms that adjust based on model confidence. Li et al. (2023a) introduced an "adaptive plausibility constraint" in their contrastive decoding framework, which filters tokens using a threshold proportional to the maximum probability: $\alpha \cdot \max_w P(w|x_{<i})$. This constraint is similar to min-p as a threshold relative to model confidence rather than absolute. The key distinction is that Li et al. (2023a) developed this as a constraint within a contrastive objective using expert and amateur models, min-p treats confidence-scaled truncation as the standalone sampling mechanism, a much simpler approach.

3 Min-p Sampling

The core idea of **min-p sampling** is to dynamically adjust the sampling threshold based on the model's confidence at each decoding step, improving sensitivity to context and uncertainty while better balancing creativity and coherence, especially at high temperatures.

3.1 Overview of Min-p Sampling

In autoregressive generation, a language model predicts the probability distribution over the vocabulary for the next token, conditioned on the existing sequence. At each step, the model selects a token from this distribution. Min-p sampling is a stochastic method that varies its truncation threshold based on the model's confidence, making the threshold context-sensitive.

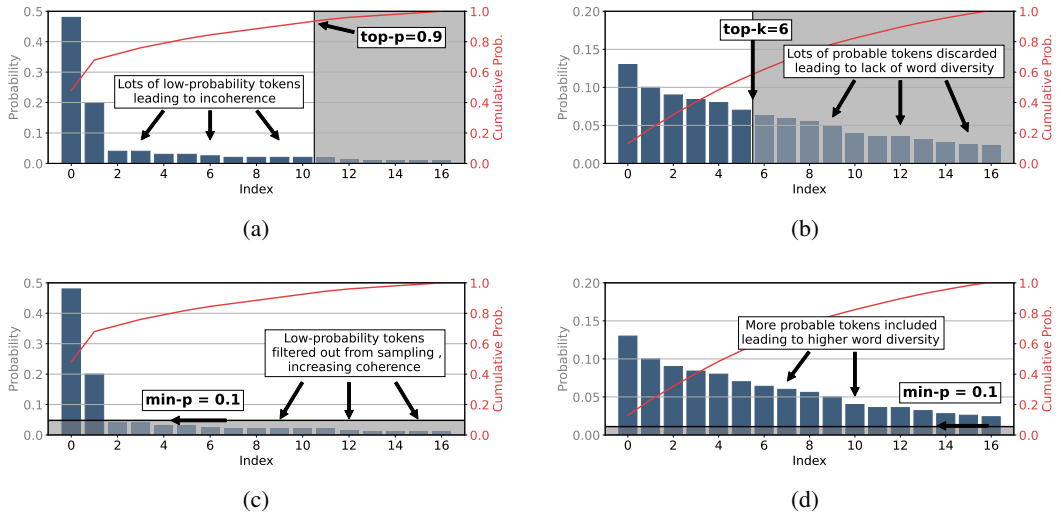

Figure 1: Comparison of sampling methods. (a) **Top-p sampling** in a high-certainty case. (b) **Top-k sampling** in a low-certainty case. (c) **Min-p sampling** in a high-certainty case. (d) **Min-p sampling** in a low-certainty case. Min-p sampling dynamically adjusts its threshold based on the model's confidence, focusing on likeliest tokens when confident and allowing more diverse options when uncertain. This dynamic threshold balances coherence and diversity better than top-p and top-k.

Formally, at each time step t, let $\mathcal{V}$ denote the vocabulary, and $P(x_t \mid x_{1:t-1})$ represent the conditional probability distribution over the vocabulary for the next token x_t. Min-p sampling works as follows:

1. **Calculate the Maximum Probability:** Identify the maximum probability token in the distribution, denoted as $p_{\max} = \max_{v \in \mathcal{V}} P(v \mid x_{1:t-1})$.
2. **Truncation Threshold:** Set a base probability threshold, $p_{\text{base}} \in (0, 1]$ Scale it by $p_{\max}$:
$$p_{\text{scaled}} = p_{\text{base}} \times p_{\max} \tag{1}$$
This threshold ensures that tokens with sufficiently high relative probabilities are considered while filtering out less probable tokens in a context-dependent manner.
3. **Define Sampling Pool:** Construct pool $\mathcal{V}_{\text{min}}$ of tokens with probabilities $\geq p_{\text{scaled}}$:
$$\mathcal{V}_{\text{min}} = \{v \in \mathcal{V} : P(v \mid x_{1:t-1}) \geq p_{\text{scaled}}\} \tag{2}$$
4. **Sample from Pool:** Sample the next token x_t from the normalized set $\mathcal{V}_{\text{min}}$:
$$P'(v) = \frac{P(v \mid x_{1:t-1})}{\sum_{v' \in \mathcal{V}_{\text{min}}} P(v' \mid x_{1:t-1})} \quad \text{for } v \in \mathcal{V}_{\text{min}} \tag{3}$$

3.2 Intuition Behind Min-p Sampling

The key intuition behind min-p sampling is that **token truncation thresholds are relative and depend on how certain the distribution is for that token**, and not absolute thresholds. When the model is highly confident about the next token (i.e., $p_{\max}$ is high), min-p restricts the pool to high-probability candidates to maintain coherence. Conversely, when the model is less confident, relaxing the sampling pool allows for a more creative and diverse generation. Unlike top-p sampling, which truncates the distribution based on a fixed cumulative probability, min-p dynamically adjusts the threshold based on the model's confidence, leading to more context-sensitive generation.

Figure 1 illustrates the effects of different sampling methods, including min-p, on token probability distributions. In subfigure (a), we show an initial probability distribution over tokens. Subfigures (b), (c), and (d) demonstrate how top-p, top-k, and min-p sampling truncate this distribution. Min-p dynamically adjusts its filtering threshold based on the model's confidence, focusing on high-probability tokens when confident and including more diverse options when uncertain. This dynamic behavior helps min-p balance coherence and diversity more effectively than top-p and top-k sampling.